

SEC P vs NP log 2026-04-29

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-16 20:13 UTC

SEC P vs NP — daily log 2026-04-29

25 cycles recorded

GF(2)-Rank Defect of Term-Indicator Matrix Lower-Bounds Monotone DNF for k-CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid theory — the GF(2) vector matroid M_F realized by the term-indicator matrix $A_F \in \mathbb{F}_2^{m \times N}$ of a monotone DNF $F = \bigvee_{i=1}^m T_i$ (rows $\{T_i\}$, columns indexed by variables). Its rank $\rho_2(F) := \text{rank}_{\mathbb{F}_2}(A_F)$ is the canonical computable matroid invariant in the Whitney-Welsh tradition; the corresponding ‘rank defect’ $\Delta(F) := \log_2(m(F)+1) - \log_2(\rho_2(F)+1)$ is a polymatroid-style quantity (subadditive log-rank vs. log-cardinality) that is rarely deployed in monotone circuit lower bounds (Razborov 1985 used Bollobás set-pair / sunflower closure, not GF(2) span of clutter indicators). $\times$ Monotone DNF representation size of the k-CLIQUE_n indicator on $N = C(n,2)$ edge variables; equivalently, the minimum-cardinality clutter of minimal terms whose union gives k-CLIQUE_n.
- **Recorded:** 2026-04-29 00:11 UTC
- **Entry ID:** 023dd98c00ee

Statement

Let F be a monotone DNF on N variables with minimal-term family $H_F = \{T_1, \dots, T_m\}$, A_F its $m \times N$ indicator matrix, and define $\Delta(F) := \log_2(m+1) - \log_2(\text{rank}_{\mathbb{F}_2}(A_F)+1)$. Conjecture: (a) $\Delta(F \wedge G) \leq \Delta(F) + \Delta(G) + 1$ and $\Delta(F \vee G) \leq \max(\Delta(F), \Delta(G)) + 1$ (approximate submodularity through Boolean operations); (b) $\Delta(F) \leq \log_2(N+1)$ for any monotone DNF with $m \leq \text{poly}(N)$ (since $\text{rank}_{\mathbb{F}_2}(A_F) \geq 1$ and $m \leq \text{poly}(N)$); (c) for the canonical k-CLIQUE_n DNF $F^{\wedge\{n,k\}}$ with $k = \lfloor n/2 \rfloor$, $\Delta(F^{\wedge\{n,k\}}) \geq n/2 - 3 \cdot \log_2 n - 4$ for all $n \geq 6$, witnessing a $\Omega(n)$ gap between log-cardinality and log-GF(2)-rank.

Rationale

The GF(2) span of clique-edge-indicator vectors collapses to a sub-quadratic-rank object ($\text{rank} \leq C(n,2)$), while the minimal-term count $C(n, \lfloor n/2 \rfloor) \approx 2^n / \sqrt{n}$ is exponential — so the LOG ratio is $\Theta(n)$, much larger than any rank deficit attainable by poly-size clutters. Razborov’s approximation method tracks subadditive measures across AND/OR gates; an honest submodular Δ that is forced to $\Omega(n)$ on the clique clutter is exactly the missing ingredient for a Razborov-style argument that does not require sunflower closures, since GF(2)-rank is robust under union operations on supports. The bridge from matroid representability to monotone DNF size is concrete (one Gaussian elimination per gate), avoiding the natural-proofs / algebrization barriers that block spectral or Fourier approaches.

Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2210.12983v2] Poisson multi-Bernoulli mixture filter with general target-generated measurements and arbitrary clutter

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 118, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 78, in run_trial
    F_random = random_poly_dnf(n, n, 2) + random_poly_dnf(n, n**2, 3) + random_poly_dnf(n, n * math.floor(n/2), 4)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 56, in random_poly_dnf
    selected_edges = random.sample(edge_set, w)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence. ")
```

TypeError: Population must be a sequence. For dicts or sets, use sorted(d).

Judge reasoning

The test crashed with a `TypeError` in `random_poly_dnf` (`random.sample` called on a non-sequence) before any data was produced, so neither the support nor falsification conditions could be evaluated. | next: Fix `random_poly_dnf` by passing a list (e.g., `list(edge_set)` or `sorted(edge_set)`) to `random.sample` and rerun the 30-seed sweep.

Coarse-Equivalence Invariance of Protocol-Pullback Multiplicity Across Bit-Permuted Gadgets

- **Verdict:** INCONCLUSIVE
- **Bridge:** Coarse Geometric Lifting (CGL) — Coarse Geometry / Roe Algebras (within Geometric Group Theory) $\times$ Deterministic 2-party communication complexity, specifically the multiplicity m_Π of the RoeCover extracted from a deterministic protocol's ProtocolPullback for the lifted function $f \circ G^n$
- **Recorded:** 2026-04-29 00:24 UTC
- **Entry ID:** aa9fe4edbf5

Statement

Let $G_1 = (X, Y, g, d)$ be a MetricGadget on small alphabet ($|X|, |Y| \leq 4$) and let G_2 be obtained from G_1 by a bi-Lipschitz bijection $\varphi: X \times Y \rightarrow X \times Y$ with distortion ≤ 2 (a coarse equivalence: φ permutes inputs while distorting d by at most a factor 2). Fix any boolean $f: \{0,1\}^n \rightarrow \{0,1\}$ with $n \leq 4$. Then for every deterministic protocol Π_1 computing $f \circ G_1^n$, there exists a deterministic protocol Π_2 computing $f \circ G_2^n$ with cost $\text{cost}(\Pi_2) \leq \text{cost}(\Pi_1) + O(n)$, and the multiplicities of their CoarsePullback covers at scale $R = \text{diam}(G)$ satisfy $m_{\{\Pi_2\}} \leq 2 \cdot m_{\{\Pi_1\}} + 1$. Equivalently, $\text{CLC}(f, G_1)$ and $\text{CLC}(f, G_2)$ agree up to a constant factor independent of f , instantiating Axiom A1

(Coarse-Lift Functoriality) at the level of the multiplicity invariant that A2 connects to communication complexity.

Rationale

This conjecture directly tests Axiom A1 (Coarse-Lift Functoriality) by asserting that the *multiplicity invariant* m_Π exposed by Axiom A2 (Protocol $\rightarrow$ Cover) is preserved under coarse equivalences of the underlying gadget. If A1 holds, a bi-Lipschitz ϕ on $X \times Y$ lifts coordinatewise to a bi-Lipschitz map on $(X \times Y)^n$ with the $d_\oplus$ metric (distortion preserved since $\square^1$ products of bi-Lipschitz maps remain bi-Lipschitz), so any Roe-style cover at scale R for the G_1 -lift can be transported to a cover at scale $\leq 2R$ for the G_2 -lift while at most doubling the multiplicity (each transported set may merge with one neighbor at the boundary, contributing the $+1$ slack). The $O(n)$ cost overhead in Π_2 reflects simulating ϕ coordinatewise. Verifying this empirically would falsify A1 if multiplicities diverge, and would strengthen confidence that CLC is genuinely a coarse invariant — a prerequisite for using it as a lower-bound technique against communication complexity.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2408.07701v2] A categorical interpretation of Morita equivalence for dynamical von Neumann algebras - [2312.08907v2] Roe algebras of coarse spaces via coarse geometric modules - [1706.00276v1] Counting coarse subsets of a countable group - [2110.02008v2] Lifted Reed-Solomon Codes and Lifted Multiplicity Codes - [1903.06084v8] Lifting coarse homotopies

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 49, in run_trial
    phi_perms = generate_bi_lipschitz_permutations()
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 35, in generate_bi_l
    if is_bi_lipschitz(phi):
       ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 24, in is_bi_lipschi
    if dist(phi[(x1, y1)], phi[(x2, y2)]) > distortion * dist((x1, y1), (x2, y2)):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 18, in dist
    return max(abs(ai - bi) for ai, bi in zip(a, b))
               ^^^^^^^^^
```

TypeError: 'int' object is not iterable

Judge reasoning

The test crashed with a `TypeError` in the bi-Lipschitz check (`dist()` called on integers instead of tuples) before any trials produced data, so neither the support nor falsification criteria can be evaluated. | next: Fix the `dist()` helper to handle the (x,y) pair representation correctly (ensure ϕ maps tuples to tuples and `dist` unpacks both arguments), then rerun the 5-seed sweep.

Persistent H_0 Component Count of Sign-Quotient Row Set Caps Real Rank

- **Verdict:** INCONCLUSIVE
- **Bridge:** Persistent homology / Vietoris-Rips topology of finite metric spaces (Vietoris 1927; Edelsbrunner-Letscher-Zomorodian 2002; Carlsson-Zomorodian 2005; Carlsson 2009). The 0-th persistent Betti number $\beta_0(\text{VR}(X, \varepsilon))$ of a finite metric space (X, d) at scale ε is the number of connected components of the ε -thresholded distance graph, computable by Kruskal/union-find in $O(|X|^2 \alpha(|X|))$. Rarely deployed in communication complexity (TDA literature is dominated by data-analysis applications; we are aware of <5 papers connecting VR persistence of communication matrices to log-rank / D^{cc}). $\times$ Real rank $\text{rk}_R(M_f)$ of the $2^k \times 2^k \pm 1$ communication matrix $M_f[x, y] = (-1)^{f(x, y)}$ of a Boolean function $f: \{0, 1\}^k \times \{0, 1\}^k \rightarrow \{0, 1\}$ — the canonical Lovász log-rank / Mehlhorn-Schmidt target for deterministic two-party communication complexity $D^{\text{cc}}(f) \geq \log_2 \text{rk}_R(M_f)$.
- **Recorded:** 2026-04-29 00:40 UTC
- **Entry ID:** bc750201aaf2

Statement

Let $X_f^{\pm} \subset \{\pm 1\}^{2^k} / \{\pm 1\}$ be the set of distinct rows of M_f modulo global sign, equipped with the projective Hamming distance $d_{\pm}(r, s) = \min(d_H(r, s), 2^k - d_H(r, s))$. Let $\tau_{\pm}(M_f) := \beta_0(\text{VR}(X_f^{\pm}, 2^{\{k-2\}}))$, the number of components of the graph linking r, s whenever $d_{\pm}(r, s) \leq 2^{\{k-2\}}$. Then for every Boolean f and every $k \in \{3, 4, 5, 6\}$, $\tau_{\pm}(M_f) \leq \text{rk}_R(M_f)$; a single (k, f) with $\tau_{\pm}(M_f) > \text{rk}_R(M_f)$ refutes.

Rationale

Rank- $r \pm 1$ matrices have all rows lying in an r -dimensional subspace of $\mathbb{R}^N$, and ± 1 vectors in such a subspace are constrained to a structured (centrally-symmetric) finite set whose pairwise distances avoid the ‘mid-Hamming’ regime around $N/2$ only by sitting in a small affine pattern; persistence at scale $N/4$ should therefore not be able to manufacture more than r distinct components after the $\pm$ -quotient. This bridges the topological ‘shape-counting’ of the row metric space to the algebraic rank-bound side of log-rank, a structural inequality complementary to spectral approaches and untouched by natural-proofs (rk_R is the property being lower-bounded, not a hardness predicate on truth tables) and untouched by algebrization (no oracle relativization is used).

Novelty

- Judge: NOVEL over 5 arXiv hits

Top hits consulted: - [LIPIcs.ESA.2024.29] A Euclidean Embedding for Computing Persistent Homology with Gaussian Kernels - [s2:609241bc94e187c943bd6d4ec6bcf0c1ababa] Topology in sensor networks - [s2:2505.16879] How high is ‘high’? Rethinking the roles of dimensionality in topological data analysis and manifold learning - [PL00009265] Harmonic Analysis, Real Approximation, and the Communication Complexity of Boolean Functions - [s2:f23504fcde864a63a36bc0be33c3a3f97d615] The Nature of Migration and Its Impact on Families in Peru

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7aad0d36.py", line 185, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7aad0d36.py", line 163, in run_trial
    M.append([Fraction(row[i]) for i in range(n)])
```

~~~~^^^

IndexError: tuple index out of range

### Judge reasoning

The test crashed with an IndexError before producing any RESULT line, so neither the support nor the falsification condition could be evaluated. Per rule 2, a crashed test yields INCONCLUSIVE. | next: Fix the row-construction bug (tuple index out of range when building M from row entries) and re-run the full  $4 \times 30 \times 6$  grid to obtain  $\tau_{\pm}$  and  $\text{rk}_R$  counts.

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## Secant Rank Lower Bound for Disjointness Communication Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic geometry of secant varieties  $\times$  Communication complexity
- **Recorded:** 2026-04-29 01:14 UTC
- **Entry ID:** 1a21dd39fb6f

### Statement

The secant rank of the communication matrix for the Disjointness problem  $\text{DISJ}_n$  is  $\Omega(n)$ , implying a  $\Omega(n)$  lower bound on randomized communication complexity.

### Rationale

Secant rank measures the minimal number of rank-1 tensors needed to approximate a matrix, directly linking to communication complexity via the log-rank conjecture. For  $\text{DISJ}_n$ , its matrix structure forces high secant rank due to combinatorial constraints, providing a novel algebraic invariant for proving lower bounds.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 0, elapsed: 0.82s

```
me': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8374.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.5, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.833333333334, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8374.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'con
RESULT: SUPPORTED mean=8373.511111111111 std=0 support_fraction=1.0
```

## Judge reasoning

The test used  $n=6$ , which is too small to establish  $\Omega(n)$  bounds. Metric values show no scaling with  $n$ , suggesting potential saturation or hidden structure in small cases. | next: Test with larger  $n$  (e.g.,  $n=128$ ) to observe scaling behavior of secant rank

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## Sign-Rank Distinguishing Tensor for Read-Twice Branching Programs

- **Verdict:** INCONCLUSIVE
- **Bridge:** discrepancy theory  $\times$  matrix rigidity
- **Recorded:** 2026-04-29 01:46 UTC
- **Entry ID:** d2ba6b1a27c2

### Statement

For any read-twice branching program  $P$  computing a Boolean function  $f$ , the sign-rank of its communication matrix  $M_P$  is  $O(\log \text{size}(P))$ , but for the inner product mod 2 function  $IP_2$ , the sign-rank of its trivial read-twice BP is  $\Omega(n)$ .

### Rationale

This conjecture leverages the connection between sign-rank (a discrepancy-theoretic invariant) and matrix rigidity (a field\_B concept). The  $IP_2$  function's exponential lower bound for read-once BPs suggests its read-twice representation must exhibit high sign-rank, while read-twice BPs for general functions should have logarithmic sign-rank due to their structured computation. This bridges the gap between BP complexity and communication complexity barriers.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 100, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 77, in run_trial
    A = generate_random_sign_matrix(n, rank)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 47, in generate_random_sign_matrix
    U = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 30, in gaussian_elimination
    factor = Fraction(A[j][i], A[i][i])
              ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
```

ZeroDivisionError: Fraction(0, 0)

## Judge reasoning

Test crashed with ZeroDivisionError during Gaussian elimination, preventing data collection | next: Debug gaussian\_elimination() to handle zero pivots and regenerate matrices

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## SOS Hierarchy Integrality Gap for Random CSPs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Sum-of-Squares hierarchy × Constraint Satisfaction Problems (CSPs)
- **Recorded:** 2026-04-29 02:17 UTC
- **Entry ID:** 90b9c49aa839

### Statement

For any constant  $\varepsilon > 0$ , there exists a CSP instance with  $n$  variables and  $m = \Theta(n)$  constraints such that the SOS hierarchy requires  $\Omega(n^{\{1-\varepsilon\}})$  levels to achieve an integrality gap of  $1 - \varepsilon$  for its relaxation.

### Rationale

This conjecture builds on the SOS hierarchy's known limitations in handling random CSPs, where the hierarchy's ability to capture exact solutions scales sublinearly with the problem size. It ties to the approach's target of analyzing refutation sizes and integrality gaps, avoiding relativization barriers by focusing on structural properties of random CSP instances.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_566ald2d.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_566ald2d.py", line 64, in run_trial
    rank = gaussian_elimination(A)
           ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_566ald2d.py", line 29, in gaussian_elim
    factor = Fraction(1, A[i][i])
             ^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
```

ZeroDivisionError: Fraction(1, 0)

## Judge reasoning

Test crashed with ZeroDivisionError, preventing data collection. Support condition cannot be evaluated without successful trials. | next: Debug gaussian\_elimination() to handle zero pivots, perhaps by adding numerical tolerance or matrix preconditioning

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## Hardness of MCSP Under Parameterized Circuit Classes

- **Verdict:** INCONCLUSIVE
- **Bridge:** meta-complexity (MCSP, K-complexity)  $\times$  hardness of approximation
- **Recorded:** 2026-04-29 02:49 UTC
- **Entry ID:** 61df9c4a8766

### Statement

MCSP is hard to approximate within any constant factor for circuits of depth-3 with polynomial-size AND, OR, and MOD<sub>p</sub> gates, implying cryptographic primitives via average-case hardness reductions.

### Rationale

This conjecture builds on Hirahara's work linking MCSP to cryptographic hardness, avoiding algebraic barriers by focusing on combinatorial circuit classes. It ties to K-complexity via parameterized approximability and aligns with Impagliazzo-Oliver's techniques for hardness amplification without natural proofs.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8dbda503.py", line 80, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8dbda503.py", line 54, in run_trial
    circuit = generate_random_circuit(n)
              ^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8dbda503.py", line 21, in generate_random_circuit
    inputs = [generate_random_circuit(n // 2, depth - 1) for _ in range(2)]
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8dbda503.py", line 21, in generate_random_circuit
    inputs = [generate_random_circuit(n // 2, depth - 1) for _ in range(2)]
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8dbda503.py", line 28, in generate_random_circuit
    circuit.append([inputs[0][i], inputs[1][i]])
                   ~~~~~~^
```

IndexError: list index out of range

### Judge reasoning

Test crashed with IndexError during circuit generation, preventing data collection. Pre-registered support condition cannot be evaluated without valid trial results. | next: Debug circuit generation recursion to handle base cases where  $n/2$  becomes zero or one

## Monotone CLIQUE Lower Bound via GF(2) Rank Defect

- **Verdict:** INCONCLUSIVE
- **Bridge:** matroid theory × polymatroid expansion
- **Recorded:** 2026-04-29 03:21 UTC
- **Entry ID:** 550befe6a1bf

### Statement

For any monotone DNF formula  $F$  representing the  $k$ -CLIQUE <sub>$n$</sub>  function, the GF(2) rank defect  $\Delta(F) = \log_2(m+1) - \log_2(\text{rank}_{\{F_2\}}(A_F))$  satisfies  $\Delta(F) \geq \Omega(n)$ , while for any DNF of size  $\text{poly}(n)$ ,  $\Delta(F) = O(\log n)$

### Rationale

The GF(2) rank defect captures the combinatorial complexity of DNF representations through matroid-theoretic lens. For  $k$ -CLIQUE, the exponential number of terms forces a large rank defect, while polynomial-size DNFs have low defect due to efficient span coverage. This measure aligns with GCT's orbit closure methods by quantifying structural rigidity.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_21558f1d.py", line 109, in <module>
 result = run_trial(seed)
 ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_21558f1d.py", line 76, in run_trial
 j = int(var[4:]) - 1
 ^^^^^^^^^^^
```

ValueError: invalid literal for int() with base 10: 'x2'

### Judge reasoning

Test crashed due to invalid literal conversion, preventing data collection to assess support fraction or counterexamples. | next: Fix the test code to handle variable parsing errors and re-run trials with multiple seeds

---

## Dual Code Distance Lower Bound for Tseitin Refutations

- **Verdict:** INCONCLUSIVE
- **Bridge:** Proof Complexity × Coding Theory
- **Recorded:** 2026-04-29 03:54 UTC
- **Entry ID:** 8b11c41c9207

### Statement

The minimal size of a resolution refutation for Tseitin formulas on expander graphs is exponentially related to the dual distance of the linear code defined by the graph's incidence matrix.

## Rationale

Tseitin formulas correspond to linear systems over  $\text{GF}(2)$ , whose solution spaces form linear codes. The dual distance of these codes governs their error-correcting capabilities. By linking refutation complexity to dual code parameters, we isolate the algebraic structure of the problem from combinatorial expansion properties, avoiding natural proofs and relativization barriers. This conjecture leverages coding-theoretic techniques (e.g., dual distance bounds) alongside proof complexity methods.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_44d298fb.py", line 141, in <module>
 result = run_trial(seed)
 ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_44d298fb.py", line 116, in run_trial
 dual_distance_value = dual_distance(A)
 ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_44d298fb.py", line 67, in dual_distance
 A_inv = matrix_inverse(A)
 ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_44d298fb.py", line 55, in matrix_inverse
 A[i][j] /= factor
```

ZeroDivisionError: division by zero

## Judge reasoning

Test crashed with division by zero error, preventing data collection. The crash suggests potential issues with matrix inversion but does not directly confirm or refute the conjecture. | next: Investigate why the incidence matrix is singular for tested expanders; verify graph construction parameters

---

## Width-5 ABPs and $\text{NC}^1$ Circuit Simulation

- **Verdict:** INCONCLUSIVE
- **Bridge:** Barrington's theorem  $\times$  Algebraic branching programs
- **Recorded:** 2026-04-29 04:23 UTC
- **Entry ID:** cd771dc62fac

## Statement

Any function computable by a depth- $d$   $\text{NC}^1$  circuit can be simulated by a width-5 algebraic branching program (ABP) of size  $O(n^d)$ , but there exist functions in  $P$  that require ABPs of exponential size to compute.

## Rationale

Barrington's theorem establishes that width-5 ABPs can simulate  $\text{NC}^1$  circuits, but the size of the ABP depends on the depth of the circuit. This conjecture extends the separation by proposing a

polynomial-size ABP for  $NC^1$  functions while suggesting exponential lower bounds for P functions, aligning with Barrington's techniques and avoiding barriers like relativization.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8438b265.py", line 181, in <module>
 result = run_trial(seed)
 ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8438b265.py", line 159, in run_trial
 p_function_size, p_function_result = run_p_function_simulation()
 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8438b265.py", line 152, in run_p_function
 abp = generate_abp_for_p_function(p_function)
 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8438b265.py", line 127, in generate_abp
 abp = [0] * (2**n + 2*n)
 ^^^^^^^^^^^^^
```

OverflowError: cannot fit 'int' into an index-sized integer

## Judge reasoning

Test crashed due to overflow error, preventing evaluation of conjecture | next: Optimize ABP generation algorithm to handle large n values

---

## AC<sup>0</sup> PARITY Lower Bound via Linear Independence Invariant

- **Verdict:** INCONCLUSIVE
- **Bridge:** communication complexity × real algebraic geometry
- **Recorded:** 2026-04-29 05:07 UTC
- **Entry ID:** 0423a578c1a6

## Statement

For any AC<sup>0</sup> circuit C computing PARITY on n inputs, the minimal number of linearly independent functions it can compute is  $\Omega(\log \text{size}(C))$

## Rationale

This invariant leverages the linearity of PARITY and the structural limitations of AC<sup>0</sup> circuits. By measuring the circuit's ability to represent linearly independent functions (a natural group action under variable permutations), we isolate the complexity inherent in computing PARITY. The logarithmic lower bound on this invariant would directly contradict the existence of small AC<sup>0</sup> circuits for PARITY.

## Novelty

- Judge: NOVEL over 6 arXiv hits

Top hits consulted: - [1409.1534v1] Algorithms in Real Algebraic Geometry: A Survey - [2506.14060v16] Linear Geometry and Algebra - [0912.2255v3] Test ideals via algebras of  $p^{-e}$ -linear maps - [1101.3737v2] Continuous rational functions on real and  $p$ -adic varieties - [1606.00596v3] Randomized Polynomial Time Identity Testing for Noncommutative Circuits

## Empirical Test

- exit code: 124, elapsed: 240.06s

TIMEOUT after 240s

## Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation time

---

## Representation-Theoretic Rank Invariant for Permanent vs Determinant

- **Verdict:** INCONCLUSIVE
- **Bridge:** Schur-Weyl duality, plethysm, algebraic combinatorics  $\times$  Algebraic complexity theory (det vs perm separation)
- **Recorded:** 2026-04-29 05:27 UTC
- **Entry ID:** 694c0e067e6e

## Statement

There exists a representation-theoretic invariant  $\rho(f)$  of homogeneous polynomials  $f$  such that  $\rho(\text{perm}_n) > \rho(\det_m^{O(1)})$  for all  $m < n^{\{1.5\}}$ , and  $\rho$  is preserved under linear substitutions.

## Rationale

This conjecture leverages Schur-Weyl duality and plethysm to define  $\rho(f)$  as the minimal number of symmetric tensors required to decompose  $f$  under the action of the general linear group. By bounding  $\rho(\text{perm}_n)$  via Young tableau multiplicities and contrasting it with the stable rank of  $\det_m^{O(1)}$ , it aligns with Landsberg's tensor rank approach while avoiding natural proofs barriers through algebraic geometry.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3ea03597.py", line 98, in <module>
 result = run_trial(seed)
 ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3ea03597.py", line 73, in run_trial
 det_rank = representation_theoretic_rank(det_m(isqrt(n ** 1.5)))
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3ea03597.py", line 68, in det_m
 return [1] * (m - 1) + [0]
```

```
TypeError: can't multiply sequence by non-int of type 'float'
```

## Judge reasoning

Test crashed due to type error in calculating m; insufficient data to evaluate conjecture | next: Fix the test code to handle integer arithmetic for m calculation

## Submodular Matroid Rank Defect for Monotone CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** matroid theory  $\times$  monotone circuit complexity
- **Recorded:** 2026-04-29 05:58 UTC
- **Entry ID:** 8c018be050c5

### Statement

There exists a submodular function  $\mu$  on monotone DNF formulas such that  $\mu(F) \leq O(\log n)$  for any DNF of size  $\text{poly}(n)$ ,  $\mu(k\text{-CLIQUE}) = \Omega(n)$ , and  $\mu(F \wedge G) \leq \mu(F) + \mu(G) - 1$  for all DNFs  $F, G$ .

## Rationale

This conjecture leverages matroid rank submodularity to quantify the structural complexity of monotone DNFs. The submodularity ensures that redundant terms in the DNF do not disproportionately inflate the measure, while the CLIQUE’s exponential term dependency forces  $\mu$  to scale with  $n$ . This aligns with Razborov’s polymatroid expansion framework but avoids algebraic geometry barriers.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.64s

```

erexample': 'Submodularity test failed'})
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold': True}
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold': True}
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold': True}
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold': True}
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold': True}
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold': True}
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold': True}
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold': True}
TRIAL: {'metric_name': 'submodularity_test', 'metric_value': None, 'instances_tested': 100, 'conjecture_hold': True}
TRIAL: {'metric_name': 'clique_test', 'metric_value': None, 'instances_tested': 40, 'conjecture_hold': True}

```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ed7ced5.py", line 103, in <module>
 first failing seed = next((r["seed"] for r in results if not r["conjecture holds"]), None)
```

File "/home/ludo/Scrivania/SEC/research/pvsnp sandbox/test 2ed7ced5.py", line 103, in &lt;genexpr&gt;

```
first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~^
```

KeyError: 'seed'

## Judge reasoning

Test crashed before collecting data; cannot verify support fraction or counterexamples | next: Fix the test to handle seed generation properly and re-run with sufficient iterations

# SOS Integrality Gap and Secant Variety Dimension for Polynomial CSPs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Geometry  $\times$  Sum-of-Squares Hierarchy
- **Recorded:** 2026-04-29 06:32 UTC
- **Entry ID:** c300a1171a3c

## Statement

The integrality gap of the Sum-of-Squares (SOS) hierarchy for a polynomial constraint satisfaction problem (CSP) is bounded by the dimension of the secant variety of the variety defined by the CSP's constraints.

## Rationale

Secant varieties, which capture the geometry of polynomial systems, provide a natural link to algebraic geometry. By connecting the SOS hierarchy's performance to the secant variety's dimension, we explore a novel interplay between geometric complexity and optimization. This conjecture avoids prior work on secant ranks in communication complexity and focuses on polynomial CSPs, leveraging algebraic geometry's tools to analyze SOS lower bounds.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

```
jecture_holds': False, 'counterexample': 'SOS gap does not match secant variety dimension'}
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'SOS integrality gap', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b9c92219.py", line 81, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b9c92219.py", line 81, in <genexpr>
 fi
```

## Judge reasoning

Test crashed before producing reliable data, preventing validation of the counterexample. | next:  
Implement error handling for missing 'seed' field and re-run with more instances

---

## Protocol-Induced Covers on Tensor-Powered Girth Gadgets Exhibit Logarithmic Multiplicity Growth

- **Verdict:** INCONCLUSIVE
- **Bridge:** Coarse Geometric Lifting (CGL) — Geometric Complexity Theory  $\times$  Communication complexity class  $CC(f \circ G^n)$
- **Recorded:** 2026-04-29 07:23 UTC
- **Entry ID:** 578b4afa80c0

### Statement

For a MetricGadget  $G$  with positive asymptotic dimension ( $\text{asdim}_R(G) \geq 1$  at scale  $R = \text{diam}(G)$ ), and its  $n$ -fold tensor power  $G^{\{\otimes k\}}$  used as a gadget in the lift  $f \circ (G^{\{\otimes k\}})^n$ , every deterministic communication protocol  $\Pi$  computing  $f \circ (G^{\{\otimes k\}})^n$  with cost  $c$  induces a ProtocolPullback that forms a RoeCover of multiplicity  $m_\Pi \geq \log_2(k+1)$  for sufficiently large  $k$ , where the scale  $R_\Pi$  depends only on  $G$ .

### Rationale

This sub-conjecture tests Axiom A3 (Asdim Amplification) in tandem with A2 (Protocol  $\rightarrow$  Cover). If  $\text{asdim}(G^{\{\otimes k\}})$  grows at least linearly in  $k$  (up to constants), then A2 implies that any protocol must induce a cover whose multiplicity is at least the logarithm of the asdim-amplified structure. The ProtocolPullback must therefore reflect this coarse geometric blowup, forcing  $m_\Pi$  to grow with  $k$ . Since CLC is meant to bound communication complexity via coarse invariants, failure of multiplicity to grow logarithmically would invalidate the amplification mechanism needed for super-logarithmic lower bounds.

### Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1903.06084v8] Lifting coarse homotopies - [1803.07099v3] Varieties of coarse spaces - [2511.17227v3] A Lifting Theorem for Hybrid Classical-Quantum Communication Complexity - [2211.07055v3] Geometric complexity theory for product-plus-power - [0608054v2] Dispersion of Mass and the Complexity of Randomized Geometric Algorithms

### Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_309b7e61.py", line 133, in <module>
 result = run_trial(seed)
 ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_309b7e61.py", line 94, in run_trial
 lifted_G = lift_function(OR, G_k)
 ^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_309b7e61.py", line 44, in lift_function
 lifted_G[i*n+j][i*n:(i+1)*n] = [f(G[i][j])]
 ^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_309b7e61.py", line 25, in OR
 return max(x)
 ^^^^^
```

TypeError: 'int' object is not iterable

### Judge reasoning

The test crashed before producing data, preventing evaluation of the conjecture. | next: Fix the test code to handle the TypeError and re-run the experiment.

---

## Real Rank Lower Bound for ACC<sup>0</sup> Circuit Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry × ACC<sup>0</sup> Circuit Complexity
- **Recorded:** 2026-04-29 07:53 UTC
- **Entry ID:** 2af1f3860e05

### Statement

The real rank of the communication matrix of a Boolean function  $f: \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}$  is  $\Omega(\log n)$  if and only if  $f$  cannot be computed by a polynomial-size ACC<sup>0</sup> circuit.

### Rationale

Williams' approach leverages algebraic rank properties of matrices to establish lower bounds. By analyzing the real rank of the communication matrix, we can derive constraints on the circuit size, extending tropical rank analysis to real semialgebraic geometry. High real rank implies structural complexity that resists ACC<sup>0</sup> computation.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 124, elapsed: 240.28s

TIMEOUT after 240s

### Judge reasoning

Test timed out before producing results; insufficient data to evaluate support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation time

---

## Free Entropy Lower Bound for Disjointness Communication Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability Theory × Communication Complexity
- **Recorded:** 2026-04-29 08:50 UTC
- **Entry ID:** 44428529acab

## Statement

The free entropy of the DISJOINTNESS communication matrix  $M_n$  is  $\Omega(n)$ , implying that its randomized communication complexity is  $\Omega(n)$ .

## Rationale

Free entropy quantifies the complexity of noncommutative distributions. For the DISJOINTNESS matrix, its high free entropy reflects the inherent structure requiring  $\Omega(n)$  communication, mirroring Forster's signed-rank technique but via free probability's noncommutative framework.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 0, elapsed: 0.03s

```
nterexample': 'non-positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 435, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 10, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 435, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 105, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 45, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 105, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 45, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 10, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 45, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 10, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': None, 'instances_tested': 190, 'conjecture_hol
positive determinant'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': 0.0, 'instances_tested': 780, 'conjecture_hol
RESULT: FALSIFIED counterexample='non-positive determinant' first_failing_seed=11
```

## Judge reasoning

The metric computation failed for non-positive determinants, invalidating free entropy measurements. The test's counterexamples suggest issues with metric definition. | next: Fix metric to handle non-positive determinants and revalidate computations

---

## Roe-Cover Multiplicity Lower Bound from Asymptotic Dimension Growth in Tensor-Powered Equality Gadgets

- Verdict: INCONCLUSIVE

- **Bridge:** Coarse Geometric Lifting (CGL) — Geometric Group Theory / Communication Complexity  $\times$  Deterministic Communication Complexity ( $D^{\{cc\}}$ )
- **Recorded:** 2026-04-29 10:34 UTC
- **Entry ID:** 3035afde3273

## Statement

For the equality gadget  $G_{eq}$  with  $|X|=|Y|=2$  and Hamming metric, and its  $n$ -fold tensor power  $G_{eq}^{\otimes k}$ , the asymptotic dimension  $asdim_R((X \times Y)^n)$  grows linearly in  $k$  for fixed scale  $R = \text{diam}(G_{eq})$ , and every deterministic protocol  $\Pi$  computing  $f \circ G_{eq}^{\otimes k}$  of cost  $c$  induces a RoeCover with multiplicity  $m_\Pi \geq asdim_R((X \times Y)^{kn}) + 1$ ; hence  $CC(f \circ G_{eq}^{\otimes k}) \geq \log_2(asdim_R((X \times Y)^{kn}) + 1)$ .

## Rationale

This sub-conjecture tests Axiom A2 (Protocol  $\rightarrow$  Cover) in conjunction with A3 (Asdim Amplification), using tensor powers of a simple gadget to force coarse geometric growth. The equality gadget has verifiable Property A and admits explicit Folner sequences; its tensor powers amplify asymptotic dimension due to coarse product structure. A2 predicts that any low-communication protocol must induce a low-multiplicity cover, but if  $asdim_R$  grows with  $k$ , then so must the minimal cover multiplicity, forcing a communication lower bound. The conjecture targets the core mechanism of CGL: lifting query complexity via coarse dimension.

## Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2506.12420v1] Deterministic Lifting Theorems for One-Way Number-on-Forehead Communication - [1803.07099v3] Varieties of coarse spaces - [1903.06084v8] Lifting coarse homotopies - [2003.01308v1] Dislocation pipe diffusion and solute segregation during the growth of metastable GeSn - [1705.00570v1] High pressure floating-zone growth of perovskite nickelate  $\text{LaNiO}_3$  single crystals

## Empirical Test

- exit code: 1, elapsed: 0.10s

```
instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'asdim_infinite'}
TRIAL: {'metric_name': 'asdim_R', 'metric_value': inf, 'instances_tested': 0, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'asdim_R', 'metric_value': inf, 'instances_tested': 0, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'asdim_R', 'metric_value': inf, 'instances_tested': 0, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'asdim_R', 'metric_value': inf, 'instances_tested': 0, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'asdim_R', 'metric_value': inf, 'instances_tested': 0, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'asdim_R', 'metric_value': inf, 'instances_tested': 0, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'asdim_R', 'metric_value': inf, 'instances_tested': 0, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'asdim_R', 'metric_value': inf, 'instances_tested': 0, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'asdim_R', 'metric_value': inf, 'instances_tested': 0, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'asdim_R', 'metric_value': inf, 'instances_tested': 0, 'conjecture_holds': Fa
TRIAL: {'metric_name': 'asdim_R', 'metric_value': inf, 'instances_tested': 0, 'conjecture_holds': Fa
--STDERR--
Traceback (most recent call last):
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2933a807.py", line 125, in <module>
 print(f"RESULT: FALSIFIED counterexample=\"m_\Pi<{asdim_value}+1\" first_failing_seed={first_failing_seed}")
 ^^^^^^^^^^^^^
NameError: name 'asdim_value' is not defined
```

## Judge reasoning

The test crashed before producing any data, and the critic challenged the conjecture. | next: Fix the test code to handle the case where `asdim_value` is not defined and re-run the test.

---

## Additive Energy of Sipser Function Bounds $\text{ACC}^0$ Circuit Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Additive Combinatorics  $\times$   $\text{ACC}^0$  Circuit Complexity
- **Recorded:** 2026-04-29 10:45 UTC
- **Entry ID:** 50301ad64fa2

### Statement

For the Sipser function on  $n$  variables, its truth-table additive energy  $E$  is  $\Omega(n^2)$ , implying that any  $\text{ACC}^0$  circuit computing it requires size  $\Omega(n^2/\varepsilon^2)$  for any  $\varepsilon > 0$ .

### Rationale

Additive energy captures the number of structured quadruples  $(a,b,c,d)$  with  $a+b=c+d$  in the truth-table. High additive energy indicates non- $\text{ACC}^0$ -computability due to the lack of low-depth linear threshold gates to handle such structured correlations, aligning with Williams' 2014 methods for  $\text{NEXP}$  vs  $\text{ACC}^0$ .

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 124, elapsed: 240.03s

TIMEOUT after 240s

## Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Extend timeout duration and re-run test with increased computational resources

---

## Protocol-Induced Covers on High-Asdim Gadget Lifts Require Exponential-in- $Q(f)$ Multiplicity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Coarse Geometric Lifting (CGL) — Geometric Complexity Theory  $\times$  Deterministic 2-Party Communication Complexity ( $D^{\{cc\}}$ )
- **Recorded:** 2026-04-29 13:06 UTC
- **Entry ID:** 52a971f74369

### Statement

For any family of MetricGadgets  $G$  with  $\text{asdim}_R(G) \geq d > 0$  at scale  $R = \text{diam}(G)$ , and for any function  $f: \{0,1\}^n \rightarrow \{0,1\}$  with query complexity  $Q(f)$ , every deterministic protocol  $\Pi$  computing  $f \circ G^n$  induces a ProtocolPullback that forms a RoeCover of multiplicity  $m_\Pi \geq 2^{c \cdot Q(f)}$  for some constant  $c > 0$  depending only on  $d$  and  $G$ .

## Rationale

This conjecture tests A2 (Protocol  $\rightarrow$  Cover) in conjunction with the target invariant's requirement that  $CC(f \cdot G^n) \geq Q(f) \cdot \log_2(\text{asdim}_R(G)+1)$ . If high asymptotic dimension forces every protocol to induce a cover with multiplicity exponentially large in  $Q(f)$ , then the log-multiplicity lower bound directly recovers the desired communication complexity scaling. It leverages the geometric rigidity implied by positive  $\text{asdim}$ : protocols of low cost induce low-multiplicity covers, which cannot exist if the space is sufficiently 'spread out' at large scales.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8cbb71d9.py", line 108, in <module>
 result = run_trial(seed)
 ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8cbb71d9.py", line 83, in run_trial
 multiplicity = compute_cover_multiplicity(pullback, R)
 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8cbb71d9.py", line 60, in compute_cover
 if pullback[i][j] > 0:
 ^^^^^^^^^^^^^
```

TypeError: '>' not supported between instances of 'list' and 'int'

## Judge reasoning

The test crashed due to a `TypeError`, indicating an issue in the `compute_cover_multiplicity` function where a list was compared to an integer. No results were produced, preventing any determination of support or falsification. | next: Debug the `compute_cover_multiplicity` function to resolve the `TypeError` by ensuring `pullback[i][j]` is an integer or comparable numeric type before attempting comparison with 0.

---

## Free Entropy Gap in Read-Twice Branching Programs for IP<sub>2</sub>

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability Theory  $\times$  Read-Twice Branching Programs
- **Recorded:** 2026-04-29 15:43 UTC
- **Entry ID:** aad5b9e6d7b1

## Statement

For any read-twice BP  $P$  computing  $\text{IP}_2$ , the free entropy  $\rho(P) = \Theta(\log n)$ , whereas the trivial read-twice BP for  $\text{IP}_2$  has  $\rho(P) = \Omega(n^2)$ .

## Rationale

Free entropy measures non-commutative randomness, capturing structural constraints in BPs. Read-twice BPs have limited state reuse, while  $\text{IP}_2$ 's inherent correlation requires exponential resources, creating a gap detectable via free probability invariants.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 124, elapsed: 240.41s

TIMEOUT after 240s

## Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run with optimized parameters for larger instance sizes

---

## Protocol-Induced Covers from Low-Cost Protocols on High-Distortion Gadget Lifts Admit Sparse Roe Decompositions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Coarse Geometric Lifting (CGL) — Geometric Complexity Theory × Communication complexity class  $CC(f \circ G^n)$
- **Recorded:** 2026-04-29 18:05 UTC
- **Entry ID:** 634c56408dd3

## Statement

For any MetricGadget  $G$  where the maximal Hilbertian distortion of  $(X \times Y, d)$  embeddings into  $(X \times Y)^n$  under  $d_{\oplus}$  grows at least as  $\Omega(\log n)$ , and for any deterministic communication protocol  $\Pi$  computing  $f \circ G^n$  with cost  $c$ , the induced ProtocolPullback forms a RoeCover of scale  $R_{\Pi} = O(\text{diam}(G))$  and multiplicity  $m_{\Pi} \leq 2^{c - \Omega(Q(f) \cdot \log \text{asdim}_R(G))}$ , assuming  $\text{asdim}_R(G) \geq 1$  for  $R = \text{diam}(G)$ .

## Rationale

This conjecture tests Axiom A2 (Protocol  $\rightarrow$  Cover) in conjunction with the geometric obstruction posed by high embedding distortion, which limits how efficiently transcripts can cover the space without incurring coarse multiplicity. High distortion implies that Euclidean (hence protocol-induced) partitions cannot align well with the intrinsic geometry of the lift, forcing higher-than-expected multiplicity in any valid RoeCover. It leverages A3 (Asdim Amplification) indirectly by requiring  $\text{asdim}_R(G)$  to be nontrivial, thereby amplifying the gap between protocol-induced covers and optimal ones. If true, it strengthens the lifting bound by tying communication cost not just to existence of low-multiplicity covers, but to their geometric realizability under distortion barriers.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp\_sandbox/test\_1ff16270.py", line 94, in <module>

```

 result = run_trial(seed)
 ^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1ff16270.py", line 72, in run_trial
 cover_multiplicity = compute_cover_multiplicity(pullback, n)
 ^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1ff16270.py", line 56, in compute_cover
 if pullback[(i, '0', '0')] + pullback[(j, '0', '0')] <= pullback[(i, '0', '0')]:
    ~~~~~^~~~~~
KeyError: (0, '0', '0')

```

### Judge reasoning

The test crashed before producing data, and the pre-registered support condition was not unambiguously met. | next: Fix the KeyError in the test code and re-run the simulations to produce valid data.

---

## Spectral Radius of Vertex Contraction Graph Controls Tseitin Resolution Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Spectral graph theory × Resolution refutation length
- **Recorded:** 2026-04-29 21:06 UTC
- **Entry ID:** ea2a3e0f2c88

### Statement

Let  $G$  be a 3-regular graph and  $T(G,c)$  the unsatisfiable Tseitin formula with odd total charge. Let  $G \downarrow$  be the vertex contraction graph where each node represents a connected component of edges incident to a satisfied vertex under a partial assignment, and edges represent shared variables. Let  $\rho(G \downarrow)$  be the spectral radius of the adjacency matrix of  $G \downarrow$ . Then the minimal resolution refutation length  $L(T(G,c) \vdash \perp)$  satisfies  $L = 2^{\Omega(\rho(G \downarrow))}$  for random 3-regular  $G$  of size  $n \leq 40$ .

### Rationale

The contraction graph  $G \downarrow$  dynamically tracks how partial assignments merge satisfied components during resolution. Its spectral radius controls the rate of information propagation across the graph: high  $\rho(G \downarrow)$  implies rapid mixing and thus high proof complexity. This bridges local clause interactions with global expansion through a tractable dynamic invariant, avoiding direct reliance on global expansion which can be fragile under local perturbations.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.12s

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c96654d4.py", line 111, in <module>
    result = run_trial(seed)
    ^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c96654d4.py", line 81, in run_trial
    graph = generate_3_regular_graph(n)

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c96654d4.py", line 32, in generate_3_re
    neighbors = random.sample([n for n in nodes if n != node], degree - 1)
File "/usr/lib/python3.12/random.py", line 430, in sample
    raise ValueError("Sample larger than population or is negative")
ValueError: Sample larger than population or is negative

```

The test crashed before producing data, and the pre-registered support condition was not unambiguously met. | next: Fix the test code to handle the ValueError and re-run the test to produce meaningful results.

- **Verdict:** BARRIER\_HIT
- **Bridge:** {'framework\_name': 'Tropical Circuit Weight Analysis (TCWA)', 'math\_branch': 'BOUNDED\_ARITHMETIC'}  $\times$  {'complexity\_theoretic\_object': 'Tropical Circuit with polynomially many phase cells'}
- **Recorded:** 2026-04-29 23:34 UTC
- **Entry ID:** 19163a75e6d2

Rejected by ALGEBRIZATION barrier

# Star Discrepancy of Clause-Vectors Lower-Bounds Resolution Width for Random 3-SAT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Discrepancy theory (star discrepancy in the sense of Roth / Schmidt / Matousek)  $\times$  Resolution refutation width  $w(\text{F} \vdash \perp)$  on random unsatisfiable 3-CNFs
- **Recorded:** 2026-04-29 23:43 UTC
- **Entry ID:** b22241903f08

## Statement

Let  $F$  be a random unsatisfiable 3-CNF over  $n$  variables at clause density 4.5. Let  $V_F \in [0,1]^{m \times n}$  be the clause-variable matrix normalized coordinate-wise. Then  $E[D^*(V_F)] = \Omega(\log n)$  implies  $w(\text{F} \vdash \perp) = \Omega(n / \log n)$ , where  $D^*$  denotes the star discrepancy of the point set induced by the rows of  $V_F$ .

## Rationale

Star discrepancy measures how uniformly distributed a finite point set is in the unit cube, a central object in quasi-Monte Carlo and geometric discrepancy (Matousek 'Geometric Discrepancy'). High discrepancy implies clustering in projections, which we hypothesize corresponds to variable-heavy clauses that force wide resolution derivations. This connects dispersion in geometric space to proof complexity via variable scoping in refutations.

## Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [9707234v2] Variational Approach to Quantum Field Theory: Gaussian Approximation and the Perturbative Expansion around It - [0407622v3] A Study of the Composition of Ultra High Energy Cosmic Rays Using the High Resolution Fly's Eye - [9311011v2] The Great Inequality In A Hamiltonian Planetary Theory - [0907.5244v1] IRSF/SIRIUS JHKs near-infrared variable star survey in the Magellanic Clouds - [1005.0979v1] Supersymmetry in Random Matrix Theory

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a0eb35a8.py", line 87, in <module>
    result = run_trial(seed)
             ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a0eb35a8.py", line 71, in run_trial
 D_star = star_discrepancy(V_F, m, n)
             ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a0eb35a8.py", line 51, in star_discrepancy
    D = max(D, abs(sum(S[i] * V_F[i][j] for i in range(m))) / math.sqrt(m))
             ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a0eb35a8.py", line 51, in <genexpr>
 D = max(D, abs(sum(S[i] * V_F[i][j] for i in range(m))) / math.sqrt(m))
             ~~~~~^~~~~~
```

IndexError: list index out of range

## Judge reasoning

Test crashed with IndexError before producing data; cannot evaluate support/falsification metrics  
| next: Debug the star\_discrepancy function's index handling in test\_a0eb35a8.py